Unadjusted, lag-adjusted, and exposure-weighted carryover specifications under differential-correlation biomarker interactions: a factorial simulation study

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August 18, 2026

Contents

Abstract

Background. Aggregated N-of-1 trials pool repeated within-patient on/off contrasts to validate predictive biomarkers. Under covariance moderation, a multivariate-normal differential-correlation architecture in which the biomarker-treatment interaction lives in a treatment-state-dependent correlation, a companion study found that carryover costs up to roughly 31% of power in relative terms. The analyst must still decide how to represent residual exposure in the linear predictor, and it is not established which representation is most robust to the ordinarily unknown pharmacokinetic decay shape.

Methods. We compare three analysis-model specifications for the drug-exposure regressor. Unadjusted is the binary on-drug indicator, ignoring carryover altogether. Lag-adjusted augments that indicator with a lagged just-off-drug term estimating carryover non-parametrically, the classical crossover device. Exposure-weighted is a continuous indicator committing to a parametric decay and half-life, the current default. We embed them in an ADEMP-structured factorial simulation[1] crossing two DGP decay forms (exponential, Weibull) with the three specifications, three trial designs, two sample sizes, and three interaction strengths (216 cells, 500 replicates each), plus four sensitivity blocks and a cluster-robust recalibration. All three specifications are fitted to the same simulated datasets, so every contrast is paired, and each performance measure is reported with its Monte Carlo standard error (MCSE).

Results. Two findings stand out. First, the lagged-treatment term buys nothing. Lag-adjusted and Unadjusted are statistically indistinguishable across all 216 cells, differing by a mean of 0.003 in power against an MCSE of 0.018, and agreeing to three decimal places on bias, mean squared error, and coverage. Because the two share an estimand and differ by exactly one nuisance term, this is a clean null. Second, Exposure-weighted attains the highest power only in the two designs with a blinded-discontinuation phase (Hybrid and OL+BDC), where its advantage is about 10 percentage points (0.860 vs 0.766 for Unadjusted at the reference cell). Under the classical crossover (CO) design it is markedly inferior (0.488 vs 0.830). Where it leads, the lead is robust to decay-shape and half-life mis-specification and to autocorrelation strength. The model-based interaction test is mildly conservative for all three (mean null rejection 0.036 to 0.039), an artifact of the working covariance that a CR2 cluster-robust standard error removes without disturbing the ranking.

Conclusions. Three points warrant emphasis. First, under the covariance-moderation architecture we recommend the exposure-weighted predictor, reported with a cluster-robust standard error, for the Hybrid and OL+BDC designs. Under a crossover design it should not be used, and the unadjusted binary indicator is materially better. Second, the classi-

cal lagged-treatment remedy cannot be recommended in any design examined here, since it never improved on ignoring carryover altogether. We trace this to its entering the model as a nuisance main effect, which leaves the attenuation of the interaction untouched. Third, across all conditions anticipated dropout governs power far more than the choice among specifications, and should dominate design effort accordingly. This manuscript restricts the grid to the covariance-moderation architecture and to the exponential and Weibull decay forms. The mean-moderation architecture and linear decay are evaluated in the archived companion drafts `archive/report.Rmd` and `archive/report-slim.Rmd`.

1 Introduction

In a single N-of-1 trial, one patient is crossed repeatedly between active treatment and a comparator, and the within-patient contrast between on-drug and off-drug periods estimates that patient's individual treatment effect. An aggregated N-of-1 trial pools many such series into a mixed model, trading some within-patient resolution for population-level inference at sample sizes considerably smaller than a parallel-group trial would require[2]. The estimand is the biomarker-treatment interaction, whether a baseline covariate B_i modifies the treatment effect. We examine three trial designs that differ in how on-drug and off-drug occasions are arranged, a classical crossover (CO), an open-label design followed by blinded discontinuation (OL+BDC), and a Hybrid N-of-1 design, each defined in Section 2.2.

The inferential complication specific to this setting is carryover. A drug's pharmacological effect does not switch off abruptly at discontinuation but decays over days or weeks, so nominally off-drug observations still carry residual exposure. An exposure regressor that ignores this is a mis-measured version of what the patient actually received, and the damage is twofold. The interaction contrast is diluted, because occasions still carrying a substantial fraction of the drug effect are coded identically to occasions carrying none, biasing the biomarker-by-exposure coefficient toward zero as any covariate measured with error would. And those same occasions are heterogeneous in a way the model does not represent, so the unexplained variation inflates the standard error of the coefficient that has just been attenuated. The test loses on both counts, and power falls accordingly.

This is a loss of power, not of validity. Under the null, $c_{bm} = 0$, no biomarker-response association exists at any level of residual exposure, so a mis-coded regressor has no interaction to distort and test size is unaffected by exposure coding. The mild conservatism observed in Section 3.5 has a separate origin in the working covariance and applies equally across specifications. Carryover mis-specification is therefore a question of efficiency, and we frame the study in terms of recovered power.

Three answers are available for how to represent residual exposure in the linear predictor, corresponding to three positions an analyst can take toward the unknown decay. The analyst may deny it, coding exposure strictly on or off, may estimate it, adding a nuisance term whose magnitude the data determine, or may assume it, committing to a decay curve and a half-life. A companion study[3], building on Hendrickson[2], shows that under the covariance-moderation architecture, in which the interaction lives in the treatment-state-dependent covariance $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, carryover erodes power directly and design-dependently. At a half-life of one week, the relative loss is roughly 31% under OL+BDC but only 3 to 5% under crossover and Hybrid, against 1 to 3% under an alternative architecture where the biomarker moderates the mean response directly. We restrict attention here to covariance moderation, where the carryover problem is largest and Hendrickson's original results were obtained. That earlier work fixed the decay shape at exponential and evaluated only one analysis specification, leaving open which representation of residual exposure performs best when the true decay shape is not known precisely, as is ordinarily the case.

We compare three specifications. Unadjusted is the binary on-drug indicator, ignoring carryover entirely and serving as the benchmark any remedy must beat. Lag-adjusted augments it with a lagged just-off-drug term that estimates carryover magnitude without assuming a functional form, following the classical crossover-trial device of Jones and Kenward[4]. Exposure-weighted is a continuous indicator committing to a parametric decay form and half-life, the current pmsimstats-ng default. Unadjusted and Lag-adjusted estimate the same coefficient and are fitted to the same simulated datasets, differing by exactly one nuisance term, so their contrast isolates that term's contribution alone. Exposure-weighted differs on a second axis as well, since it changes the regressor carrying the interaction. We cross all three against exponential and Weibull data-generating decay forms and ask whether one can be recommended outright, or whether the choice depends on trial design, following the ADEMP reporting framework of Morris, White, and Crowther[1] throughout. Section 2 describes the simulation design in ADEMP order, Section 3 gives the results, and Section 4 takes up the practical implications.

2 Simulation design

The study simulates complete aggregated N-of-1 trials and analyzes each one under every specification. A single replicate proceeds as follows. Patients are allocated across the randomization paths of the chosen trial design (Section 2.2). For each patient we draw a pre-treatment biomarker together with the three latent response components, namely the pharmacological response, the natural-history trend and the placebo-belief component, jointly across all eight measurement occasions from a multivariate normal distribution whose means follow modified Gompertz trajectories and whose covariance carries within-component AR(1) autocorrelation, cross-component correlations, and the treatment-state-dependent biomarker-response correlation that encodes the interaction (Section 2.3). The observed symptom score at each occasion is the patient's baseline severity less the sum of the three components, so that a beneficial effect lowers the score. Carryover enters through a decay function of time since discontinuation, which keeps part of the pharmacological effect present at nominally off-drug occasions. All three analysis specifications (Section 2.4) are then fitted to that same simulated dataset, and we record the interaction coefficient, its standard error and its p-value from each. This is repeated 500 times for every parameter combination in the grid of Section 2.5.

Fitting all specifications to a common dataset is deliberate. It makes every contrast among them paired, so that differences are estimated with substantially less Monte Carlo variance than independent draws would give. Throughout, N denotes the **total** number of patients in a trial, allocated as evenly as possible across the design's randomization paths, so a design with P paths assigns about N/P patients to each. At $N = 70$ this is 35 per path under the two-path designs and 17 or 18 per path under the four-path Hybrid design.

2.1 Aims

The aim is to quantify, under covariance moderation, the cost of mis-specifying the carryover representation in the analysis model, and to determine which of the three candidate strategies performs better across the grid of data-generating decay forms and trial designs. A second aim, addressed in the Tier 2 sensitivity blocks of Section 3.6, is to check whether the resulting ranking survives perturbation of within-subject autocorrelation, dropout rate, half-life accuracy, and interaction size.

2.2 Trial designs

The three designs differ in how on-drug and off-drug occasions are arranged, and so in how much post-discontinuation information each generates. All three schedule eight measurement occasions per patient across randomization paths, each a fixed on-drug/off-drug schedule, and each design carries an expectancy weight (one during open-label periods, one half during blinded periods) scaling the placebo-belief component.

Crossover (CO). A traditional two-period within-patient crossover, blinded throughout, occasions spaced 2.5 weeks apart from 2.5 to 20 weeks. One path is active for the first four occasions then comparator, and the other reverses that order and so never discontinues, contributing no post-discontinuation observations.

Open-label followed by blinded discontinuation (OL+BDC). Open-label titration at 4, 8, 12, and 16 weeks, then weekly follow-up to week 20 with a blinded late switch to placebo. The two paths discontinue after week 17 or 18, so only two or three occasions follow.

Hybrid. The Hendrickson design and this program's reference. Open-label titration at weeks 4 and 8 is followed by blinded discontinuation weekly from week 9 to 12, then a brief two-occasion crossover at weeks 16 and 20. The four paths vary discontinuation timing and crossover-arm order. It is the richest of the three in on-off transitions and the only one allowing more than one discontinuation per patient.

One consequence bears directly on the results. The first post-discontinuation occasion falls one week after switching under OL+BDC and Hybrid, but 2.5 weeks under CO. At the half-lives examined here, roughly half the pharmacological effect therefore persists into the first off-drug occasion under the two non-crossover designs, against less than a fifth under CO, so the designs differ substantially in how much carryover contamination they can accumulate at all (Section 4.2).

2.3 Data-generating mechanism

Under covariance moderation, the biomarker and the biological response are drawn jointly from a multivariate normal distribution with a treatment-state-dependent correlation, $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, where ϕ is the decay form and t_{sd} the time since drug discontinuation. Mean moderation and the combined dual-channel architecture fall outside this manuscript's scope. The full cross-architecture comparison is in the archived full-scope drafts (`archive/report.Rmd`, `archive/report-slim.Rmd`). The decay function $\phi : \mathbb{R}_+ \to [0,1]$ maps t_{sd} to a residual-effect multiplier, and we evaluate two forms here, each matched to a common half-life $t_{1/2}$. A third, linear decay, appears in the full-scope manuscripts but not here.

2.3.1 Exponential decay

$$\phi_{\exp}(t_{sd}) = \exp\left(-\lambda t_{sd}\right), \qquad \lambda = \frac{\ln 2}{t_{1/2}}$$

This is the decay form used throughout the companion manuscript and implemented in `functions.R` under `implementations/tidyverse/`. It corresponds to classical first-order elimination kinetics, the default pharmacokinetic assumption, and serves as the reference against which Weibull is compared.

2.3.2 Weibull decay

$$\phi_{\text{weib}}(t_{sd}) = \exp\left(-\left(\lambda_w t_{sd}\right)^k\right), \qquad \lambda_w = \frac{(\ln 2)^{1/k}}{t_{1/2}}$$

A shape parameter k governs whether the elimination tail is heavier than exponential ($k < 1$, slower elimination) or lighter ($k > 1$, faster clearance), recovering the exponential form exactly at $k = 1$. We report $k = 0.7$ (a prolonged low-concentration tail) and $k = 1.5$ (faster clearance once concentration falls below some threshold), which with $k = 1.0$ (numerically indistinguishable from exponential) gives four decay-shape levels in the grid.

2.4 Estimand and analysis specifications

The interaction estimand is the on-drug biomarker-response correlation c_{bm}, a dimensionless quantity. Bias and empirical SE are reported on a calibrated effect-size scale (the expected difference in on-drug response per SD of the biomarker at $t_{1/2} = 0$, per `docs/19-mean-moderation-implementation-notes.tex`). Power and Type I error are on the native test-statistic scale, with the null estimand $c_{bm} = 0$. The analysis model is `nlme::lme(Sx ~ bm * X + t, random = ~1 | ptID, correlation = corCAR1(form = ~ t | ptID))` for all three specifications, which differ only in how X is constructed and whether a nuisance term is added. Unadjusted and Lag-adjusted both test the coefficient on bm:D_{it}, so every performance measure is comparable between them. Exposure-weighted tests bm:$D_{bc,it}$, a different quantity. Power and Type I error remain comparable across all three, since they are properties of the test, but bias, MSE, and coverage are not comparable across that divide (Section 3.1).

2.4.1 Unadjusted: binary on-drug indicator

$X = D_{it}$, the binary on-drug indicator (one on drug, zero off). Carryover is not represented at all, so an off-drug occasion still carrying most of the pharmacological effect is coded identically to one carrying none. This is the default an analyst obtains without considering carryover, and serves as the benchmark the other two must justify their additional structure against. It consumes no half-life and so is invariant to the analyst's pharmacokinetic assumptions.

2.4.2 Lag-adjusted: binary indicator plus estimated carryover term

$X = D_{it}$ is retained and the model is augmented with a lagged-treatment indicator $L_{it} = D_{i,t-1}(1 - D_{it})$, marking off-drug timepoints immediately following an on-drug one. Carryover magnitude is estimated from the data rather than assumed, so, like Unadjusted, this specification is invariant to the assumed half-life, following the classical crossover-trial device of Jones and Kenward[4]. Critically, L_{it} enters only as a nuisance main effect, with no bm:L_{it} product, so the interaction tested remains bm:D_{it}. Lag-adjusted can absorb mean-level carryover but cannot repair the interaction's attenuation (Section 1). It is best understood as Unadjusted with a carryover nuisance control added, not a fundamentally different model of the interaction (Section 4.1).

2.4.3 Exposure-weighted: continuous decayed predictor

$X = \mathrm{Dbc}_{it}$, a continuous exposure-decayed predictor matched to the analyst's assumed decay form and half-life, the current pmsimstats-ng default. At on-drug timepoints $D_{bc,it} = 1$. Off drug it decays as $D_{bc,it} = (1/2)^{t_{sd,it}/\hat{t}_{1/2}}$, losing half its value every $\hat{t}_{1/2}$ weeks after discontinuation. When the assumed decay form matches the true one, the predictor is well matched. When it does not, it is mis-specified in a way probed directly in block S2. The half-life assumption enters only through the predictor values handed to `lme`, not the formula, so Exposure-weighted is in effect a different model for every assumed half-life. It is the only one of the three that consumes a half-life at all, which is what the S2 sensitivity contrast (Section 3.6) is designed to probe. A limiting relationship connects the first and third. As $\hat{t}_{1/2} \to 0$, $D_{bc,it} \to D_{it}$, so Unadjusted is exactly the boundary member of the Exposure-weighted family at zero assumed half-life, and the S2 sweep is therefore a traverse anchored at Unadjusted rather than an independent check.

2.5 Factorial grid and parameter settings

The principal comparison crosses DGP decay form (exponential and Weibull at $k \in \{0.7, 1.0, 1.5\}$) against the three analysis specifications, under covariance moderation only, with the exponential row analyzed under Exposure-weighted as the matched reference cell.

Parameter	Values
Biomarker moderation (c_{bm})	0.0, 0.30, 0.45
Carryover half-life ($t_{1/2}$)	0.0, 0.5, 1.0 weeks
Weibull shape (k)	0.7, 1.0, 1.5 (Weibull form only)
Sample size (N)	35, 70
Trial design	CO, N-of-1 (Hybrid), OL+BDC
DGP architecture	B (MVN differential correlation) only
Replicates per cell	500 (target), 50 for development

The four decay-shape levels (exponential and three Weibull shapes) against three specifications, three designs, two sample sizes, and three biomarker levels give $4 \times 3 \times 3 \times 2 \times 3 = 216$ cells. The $c_{bm} = 0$ row is the Type I error check against nominal 5%, and the $t_{1/2} = 0$ row, with no carryover, gives a best-case power reference at each combination of N, design, and c_{bm} against which the cost of nonzero carryover can be read off directly.

2.6 Inference and the cluster-robust correction

The interaction test described so far is model-based. The standard error of $\beta_{bm:X}$ is read from the fitted `lme` object, valid only if the assumed residual covariance is correct. That assumption is not innocuous here. The working covariance combines a patient random intercept with a `corCAR1` AR(1) residual correlation, and the companion calibration study[5] shows this does not match the covariance the data-generating mechanism actually induces, inflating the standard error by roughly six to ten percent and depressing the rejection rate correspondingly. We quantify this with the calibration ratio κ, the mean estimated SE divided by the empirical SD of the interaction estimate across replicates ($\kappa = 1$ is correctly calibrated, $\kappa > 1$ conservative). At the null cell κ runs 1.05 to 1.10 across specifications, with empirical Type I error 0.029 to 0.039 against nominal 0.05.

The remedy is a cluster-robust (sandwich) variance estimator, which derives the variance of $\hat{\beta}_{bm:X}$ from observed between-cluster variation rather than the assumed covariance, and so remains valid under mis-specification. Clusters are patients (eight occasions each, unique across randomization paths). Because the naive sandwich form is markedly downward-biased with few clusters ($N = 35$ or 70), we use the CR2 bias-reduced estimator of Bell and McCaffrey, paired with Satterthwaite degrees of freedom (both via `clubSandwich`, replacing only the standard error, reference distribution, and p-value, leaving point estimates untouched).

We apply the correction in two places. A null-cell diagnostic ($c_{bm} = 0$, exponential DGP, $t_{1/2} = 1.0$ weeks, Hybrid, $N = 70$, 3,000 replicates) establishes whether CR2 restores nominal size (Section 3.5). A five-cell block, the reference cell plus four stress cells (small N, high autocorrelation, half-life mis-specification, 30% MCAR dropout, 500 replicates each, labeled S6 in this manuscript's block numbering), then asks whether restoring size disturbs the ranking (same section). Blocks S1 through S4 are the Tier 2 sensitivity analyses of Section 3.6. Block S5, an exploratory autocorrelation-by-carryover interaction, is outside the present scope and reported only in the archived full-scope drafts.

2.7 Performance measures and computing environment

For each cell we report Power, Type I error, Bias, Empirical SE (calibrated scale), Coverage of the nominal 95% Wald CI, MSE, non-convergence rate, and the MCSE of each, following Morris et al.[1, Tables 5-6]. We chose $n_{sim} = 500$ so the MCSE on power at $p = 0.5$ would not exceed 0.022. No simulation runs for this manuscript. We filter the same 540-cell production grid summary used in the archived full-scope drafts down to the 216-cell slice defined by covariance moderation and the two non-linear decay forms. All three specifications are fit to the same simulated dataset within each cell (common random numbers), so every pairwise contrast, especially Lag-adjusted minus Unadjusted, carries reduced Monte Carlo variance relative to independent draws. The Tier 2 sensitivity blocks and S6 recalibration are filtered from the same shared production outputs. Exact file paths and versions are given in the Reproducibility section.

3 Results

All results reported below derive from the same 500-replicate production run used in the full-scope manuscripts, filtered to the covariance-moderation architecture and the exponential and Weibull decay forms. The Monte Carlo standard error on power does not exceed 0.022 at $p = 0.5$, and differences smaller than this should not be over-interpreted. Coverage of the nominal 95% Wald confidence interval is reported for the reference cell, and non-convergence rates are reported per cell in the tables that follow.

3.1 Headline performance table

Table 1 reports performance at the reference cell (exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$), where non-convergence was zero throughout. The bias, MSE, and coverage columns are computed against Exposure-weighted's calibrated target, $\theta_{\text{true}} = -c_{bm}\sigma_{BR} = -3.6$, while Unadjusted and Lag-adjusted estimate a different coefficient, bm:D_{it}. Their apparent bias and under-coverage reflect that estimand mismatch, not a defect in the estimators, so those columns are comparable within the Unadjusted/Lag-adjusted pair but not across to Exposure-weighted. For the three-way comparison we rely on power and Type I error, properties of the test rather than the coefficient tested.

3.2 Tier 1 grid overview

Figures 1 and 2 give a full-grid view of the Tier 1 results at the exponential DGP, in the annotated-heatmap style of Hendrickson et al.[2(fig4)], with trial design as column facets, N as row-block facets, and a blue (low power) to red (high power) fill scale. Hendrickson's row axis within each N block was a set of censoring parameters; there is no censoring axis in this manuscript's grid, so it is replaced with analysis specification, the axis this manuscript's grid varies instead.

Two features are visible at a glance that the headline table and line plots elsewhere in this section make only cell-by-cell. First, within every design x N block, the Unadjusted and Lag-

Table 1: Headline performance at the reference cell (exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $n_{sim} = 500$). Power is the rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$. Bias is on the calibrated effect-size scale, $\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR} = -3.6$ (per docs/19 calibration). MCSEs are reported per Morris et al. (2019, Table 6).

t1/2	Specification	Power (MCSE)	Bias (MCSE)	Empirical SE	Coverage (MCSE)	MSE (MCSE)	Non-conv
0.0	Unadjusted	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	Lag-adjusted	0.990 (0.004)	-0.025 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	Exposure-weighted	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.5	Unadjusted	0.918 (0.012)	+0.586 (0.036)	0.807	0.938 (0.011)	0.995 (0.058)	0.000
0.5	Lag-adjusted	0.920 (0.012)	+0.583 (0.036)	0.812	0.938 (0.011)	1.000 (0.059)	0.000
0.5	Exposure-weighted	0.948 (0.010)	+0.072 (0.041)	0.907	0.972 (0.007)	0.828 (0.057)	0.000
1.0	Unadjusted	0.766 (0.019)	+1.093 (0.038)	0.847	0.828 (0.017)	1.911 (0.093)	0.000
1.0	Lag-adjusted	0.770 (0.019)	+1.099 (0.038)	0.853	0.818 (0.017)	1.935 (0.094)	0.000
1.0	Exposure-weighted	0.860 (0.016)	-0.081 (0.049)	1.085	0.968 (0.008)	1.184 (0.077)	0.000

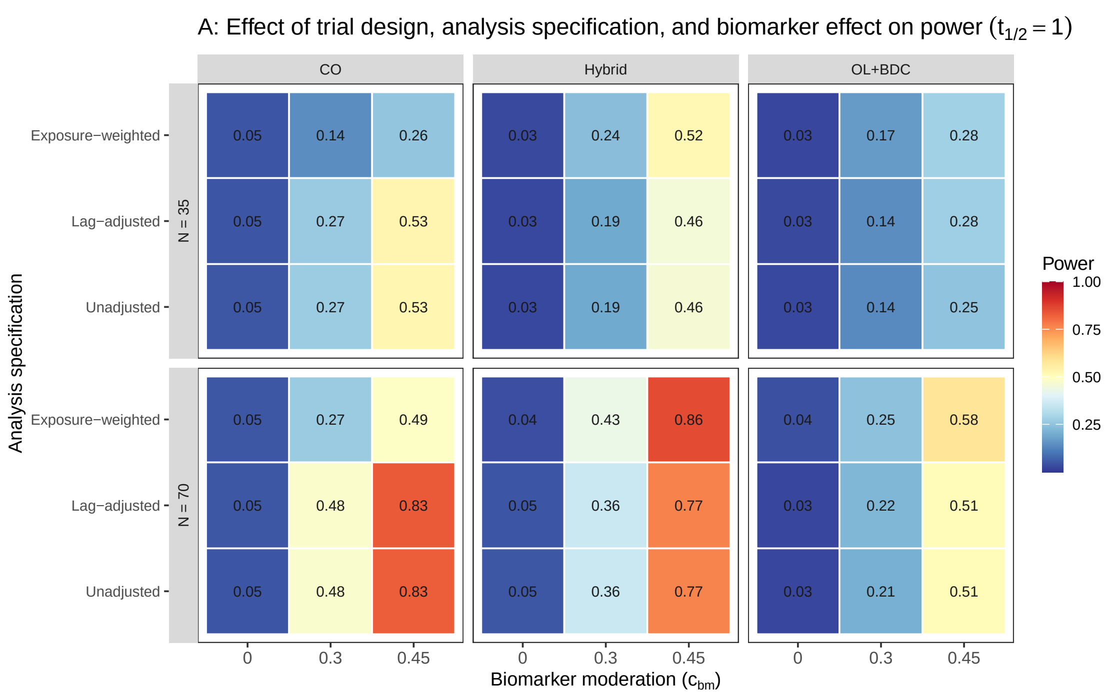

Figure 1: Tier 1 grid, biomarker-effect axis: power by trial design (columns), N (row blocks), and analysis specification (rows within block), across $c_{bm} \in \{0, 0.30, 0.45\}$ at $t_{1/2} = 1.0$ weeks, exponential DGP.

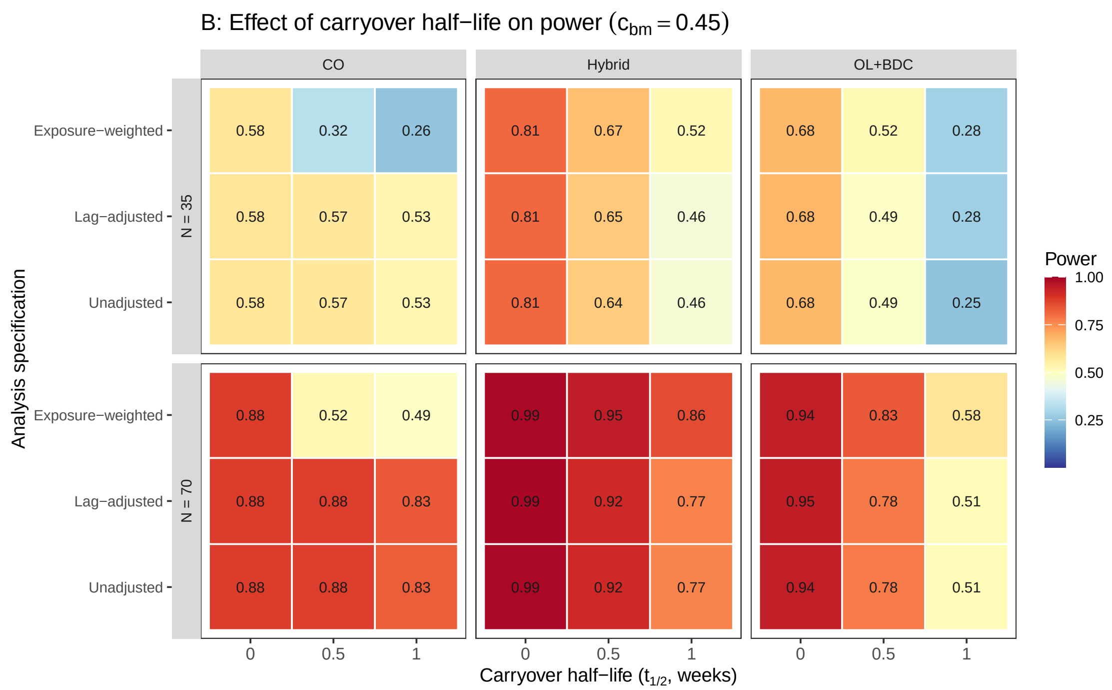

Figure 2: Tier 1 grid, carryover-axis: power by trial design (columns), N (row blocks), and analysis specification (rows within block), across $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks at $c_{bm} = 0.45$, exponential DGP.

adjusted rows are numerically identical to two decimal places at every biomarker level and half-life, the clean null of Section 3.3 displayed across the entire grid at once rather than at one reference cell. Second, the Exposure-weighted row visibly separates from the other two only in the Hybrid and OL+BDC columns, and only once carryover is present (Figure 2); in the CO column it is weaker than the other two at every half-life past zero, the design-dependence of Section 3.4 read directly off the grid.

3.3 Matched-decay baseline

At the matched cell (exponential DGP, exposure-weighted analysis, $N = 70$, $c_{bm} = 0.45$, Hybrid design), power declines from 0.988 ± 0.005 at $t_{1/2} = 0$ to 0.948 ± 0.010 at $t_{1/2} = 0.5$ to 0.860 ± 0.016 at $t_{1/2} = 1.0$ weeks (point estimate plus or minus MCSE, $n_{sim} = 500$). All 500 replicates converged at every half-life in this cell.

We had previously read the value at $t_{1/2} = 1.0$ as agreeing with the power of 0.82 reported by Hendrickson et al. for a comparable cell[2]. Unfortunately, we must withdraw that comparison. In the reference implementation, the analysis-side and data-generating carryover parameters are set independently, and the drivers that reproduce the original figures supply a half-life to the data-generating step only, leaving the analysis-side half-life at its default of zero, at which the exposure-decayed predictor collapses onto the binary on-drug indicator. The reference analysis is therefore Unadjusted, not Exposure-weighted, so the appropriate comparator is our unadjusted power at the same cell, 0.766, and the grids differ further on interaction strength and half-life, so "the same cell" was approximate in any case. We have not resolved what the published figure was computed under, and make no claim of agreement or disagreement here. Nothing else in this manuscript depends on that comparison, since the three specifications are compared against one another within a single simulation under common random numbers.

3.4 Power by analysis specification

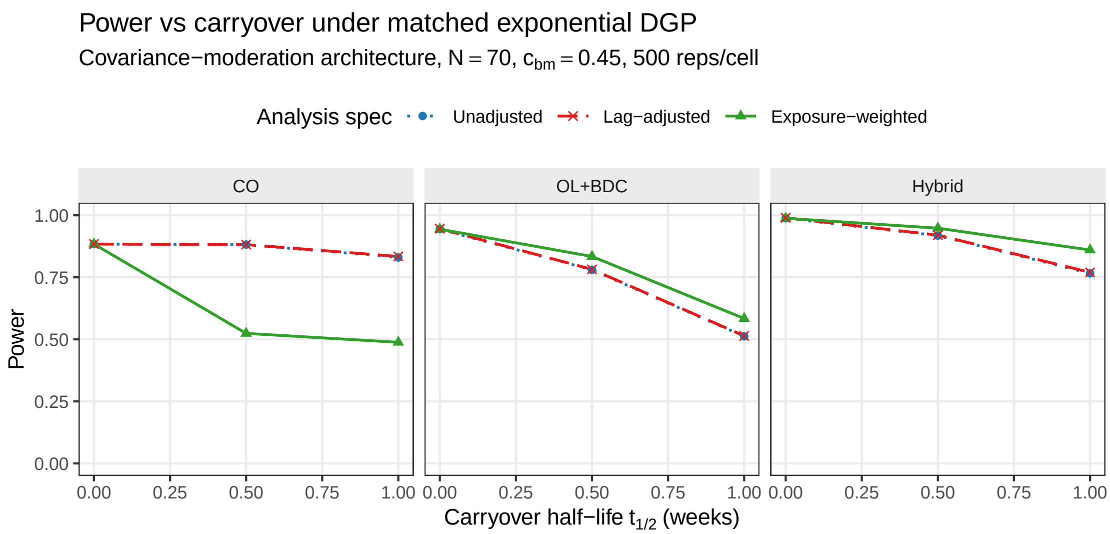

Figure 3: Power versus carryover half-life under the exponential DGP, by analysis specification and trial design.

Two features of Figure 3 carry the substance of this study. First, the Unadjusted and Lag-adjusted curves lie on top of one another in every panel and at every half-life. At the reference cell they give 0.766 and 0.770 at $t_{1/2} = 1.0$ weeks, and across all 216 cells the mean difference is 0.003, with mean bias, MSE, and coverage agreeing to three decimal places. Since the two share an estimand, are fitted to the same simulated datasets, and differ by exactly one nuisance term, this is as clean a null as the design can produce. The lagged-treatment term contributes nothing on any measure (Section 4.1). Second, the three specifications give nearly identical power at $t_{1/2} = 0$, where there is no carryover to model, and diverge as the half-life increases, with Exposure-weighted separating from the other two. It retains a modest but consistent advantage in the Hybrid and OL+BDC designs from $t_{1/2} = 0.5$ weeks onward, but not under the crossover design, where it is markedly inferior to both alternatives (0.488 against 0.830 for Unadjusted). This appears to be because the within-subject correlation in a crossover trial already carries most of the signal the decaying predictor targets, so it gains little while its down-weighting of off-drug observations works against it (Section 4).

3.5 Decay-form mis-specification

The heatmap cell at the intersection of the exponential DGP row and the Exposure-weighted column is the matched reference, with power 0.860±0.016. The Unadjusted and Lag-adjusted columns, invariant to the assumed decay form by construction, are near-identical throughout,

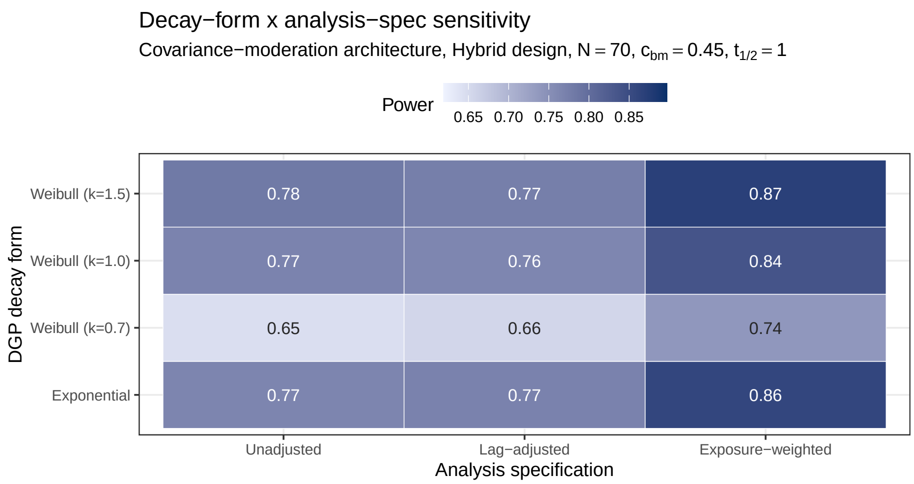

Figure 4: Power at the critical cell ($t_{1/2} = 1.0$ weeks, Hybrid, $N = 70$, $c_{bm} = 0.45$) across DGP decay forms (rows) and specifications (columns). Fill scale is stretched to the observed range, not [0,1], and not comparable to other figures.

as in Figure 3, so variation down those columns is data-generating decay and Monte Carlo noise. The remaining Exposure-weighted rows give the cost of assuming exponential decay when the truth is Weibull. Even at the heaviest-tailed form examined ($k = 0.7$), its advantage narrows but does not reverse (Section 4).

3.6 Type I error control and cluster-robust recalibration

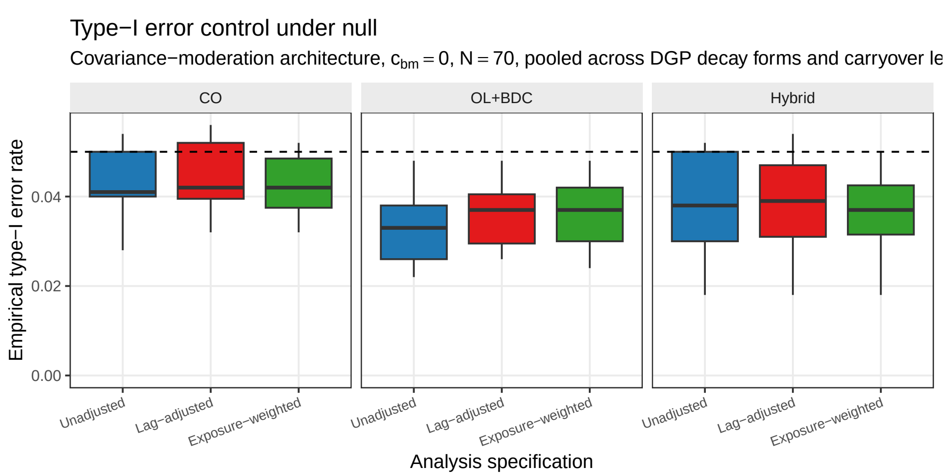

Figure 5: Empirical Type I error under the null, pooled across decay forms and half-lives, by design and specification ($N = 70$, dashed line marks nominal 0.05).

Across the null cells, empirical Type I error sits uniformly below the nominal 0.05 level for all three specifications and designs (design-level means 0.032 to 0.041). This is not sampling noise but the conservatism of the model-based test (Section 2.6), from a working covariance that does not match the data-generating one. Because the conservatism is uniform across specifications, it does not disturb the comparison among them, and the block S6 recalibration

Table 2: Conservatism of the model-based interaction test and its cluster-robust correction, at the null cell ($c_{bm} = 0$, exponential DGP, $t_{1/2} = 1.0$ weeks, Hybrid design, $N = 70$, 3,000 replicates). κ is the calibration ratio defined in Section 2.6. $\kappa > 1$ indicates a conservative test. All three specifications are conservative under the model-based test, most so Exposure-weighted ($\kappa = 1.10$, Type I = 0.029). The CR2 bias-reduced cluster-robust standard error restores κ to near one and Type I to nominal (0.045 to 0.049) for all three.

Specification	κ model	Type I model	κ CR2	Type I CR2
Unadjusted	1.06	0.036	0.98	0.048
Lag-adjusted	1.05	0.039	0.98	0.049
Exposure-weighted	1.10	0.029	1.00	0.045

Table 3: Cluster-robust recalibration at the reference cell and four stress cells (500 replicates, $c_{bm} = 0.45$). EW is Exposure-weighted, LA is Lag-adjusted, U is Unadjusted. κ is the calibration ratio of Section 2.6, here for Exposure-weighted. A value above one is a conservative test. The two gap pairs give each alternative against the unadjusted benchmark under each inference. CR2 df is the mean Satterthwaite degrees of freedom for the cluster-robust test on Exposure-weighted. Computed with `clubSandwich`.

Cell	κ model	κ CR2	EW model	EW CR2	EW−U model	EW−U CR2	LA−U model	LA−U CR2	CR2 df
Reference, $N = 70$	1.15	0.99	0.829	0.863	+0.080	+0.092	-0.003	+0.002	23
Small $N = 35$	1.13	0.96	0.558	0.590	+0.066	+0.086	-0.002	+0.000	12
High autocorr, $\rho = 0.9$	1.26	0.94	0.954	0.980	+0.052	+0.032	+0.000	-0.004	23
Half-life mis-spec	1.14	1.00	0.759	0.795	+0.010	+0.024	-0.003	+0.002	23
30% MCAR dropout	0.99	0.85	0.148	0.126	+0.010	+0.010	-0.004	+0.010	5

below confirms that a CR2 cluster-robust standard error restores nominal rejection without changing which specification leads. This also establishes that the null result of Section 3.3 is not an artifact of differential test size, since Unadjusted and Lag-adjusted are conservative to the same degree. Table 2 makes this explicit at the null cell.

Figure 6 confirms this directly. The model-based rejection rate sits below nominal across all five S6 cells and all three specifications, while CR2 restores it to within the Monte Carlo band around 0.05, and in the information-rich cells the calibration ratio κ runs from 1.07 to 1.26 under the model-based test, restored to about one under CR2, recovering two to five points of power. Table 3 reports both alternatives against the unadjusted benchmark. The Exposure-weighted advantage is held or slightly widened under CR2 at the reference cell (+0.080 to +0.092), the small-N cell (+0.066 to +0.086), and the half-life mis-specification cell (+0.010 to +0.024). It narrows under high autocorrelation (+0.052 to +0.032, though it still leads), and is small under 30% MCAR dropout (+0.010), which we read as the S3 finding restated, since dropout removes the off-drug information the advantage depends on. The Lag-adjusted gap, by contrast, is within ± 0.010 of zero at every cell under either inference, corroborating the Section 3.3 null under a second procedure and at four stress cells the main grid does not cover. This recalibration was validated only in the Hybrid, $N = 70$ cells, and should not be extended to the crossover design, where Exposure-weighted does not lead in the first place.

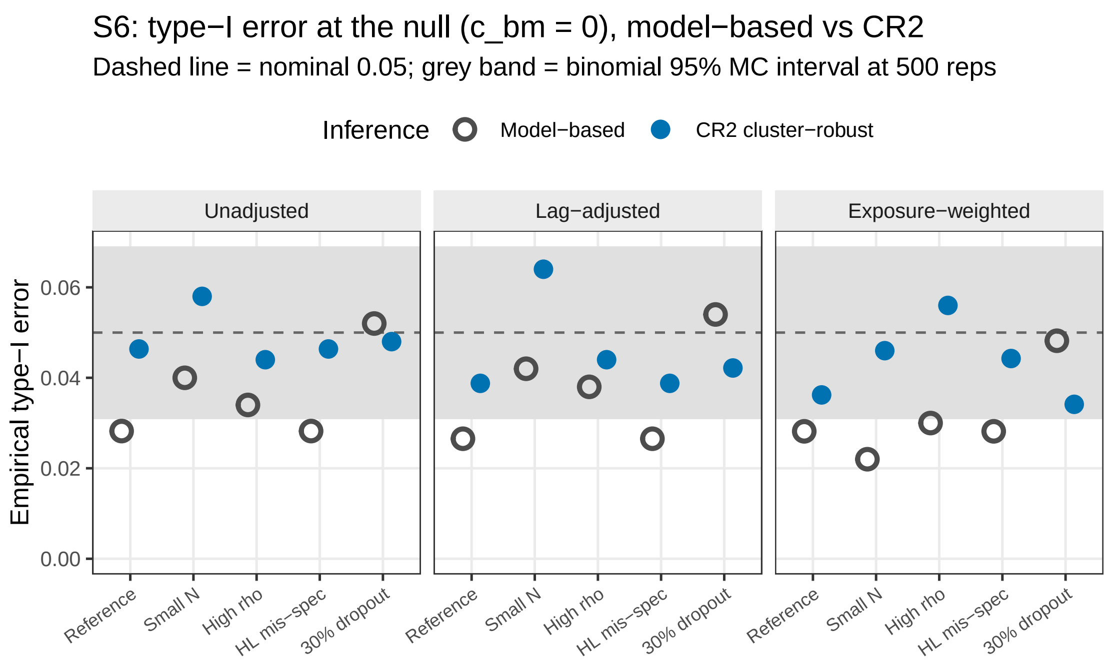

Figure 6: Empirical Type I error at the null ($c_{bm} = 0$) under model-based and CR2 cluster-robust inference, across the five S6 stress cells and the three specifications. The model-based test (open) is conservative (near 0.03), and CR2 (filled) is restored to the nominal 0.05 (dashed), within the binomial Monte Carlo band (gray). 500 null replicates per cell.

3.7 Sensitivity analyses

Six Tier 2 blocks (S1-S4, S7, S8) perturb one axis at a time around a common reference cell (Hybrid, $N = 70$, $c_{bm} = 0.45$, exponential DGP, $t_{1/2} = 1.0$), holding the remaining factors fixed; a seventh, S6, applies the cluster-robust correction of Section 3.5 at that same cell plus four stress cells, and an eighth, S5, is exploratory and reported only in the archived full-scope drafts. The blocks are reported separately here and are not pooled with Tier 1 or with each other.

3.7.1 S1: within-factor autocorrelation

All three specifications gain power monotonically with ρ, from 0.798 ± 0.018 (Exposure-weighted) and 0.738 ± 0.020 (the other two, identical to three decimals) at $\rho = 0.5$, to 0.956 ± 0.009 against 0.906 to 0.910 at $\rho = 0.9$. The Exposure-weighted advantage, roughly 0.05 to 0.08 in absolute power, is preserved at every autocorrelation level examined, suggesting autocorrelation strength and specification choice act additively rather than interacting. Lag-adjusted tracks Unadjusted throughout, never differing by more than 0.008.

3.7.2 S2: analyst-truth half-life mismatch

Block S2 crosses two true half-lives ($t_{1/2} \in \{0.5, 1.0\}$ weeks) with four analyst-assumed values ($\hat{t}_{1/2} \in \{0.25, 0.5, 1.0, 2.0\}$ weeks), giving eight cells. Only Exposure-weighted consumes $\hat{t}_{1/2}$,

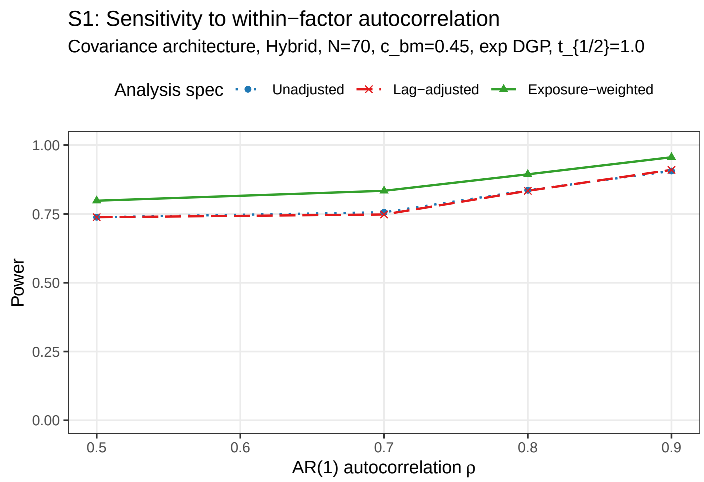

Figure 7: Power by analysis specification across AR(1) autocorrelation $\rho \in \{0.5, 0.7, 0.8, 0.9\}$.

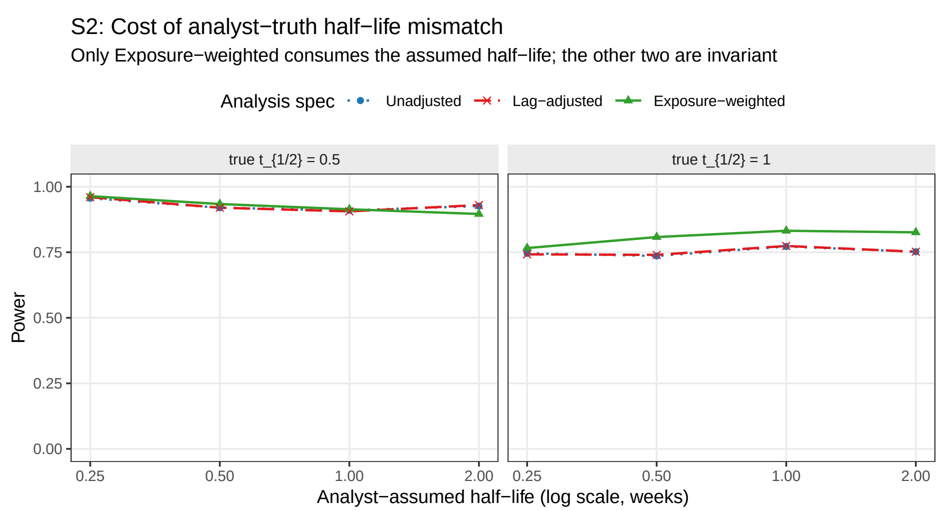

Figure 8: Power as the analyst's assumed half-life departs from the true DGP half-life. Only Exposure-weighted consumes the assumed half-life. Unadjusted and Lag-adjusted are invariant to it by construction and appear as flat reference lines.

so the cost of a wrong assumption must be read within a true half-life rather than pooled across the block.

The penalty is modest. At a true half-life of 1.0 weeks, Exposure-weighted attains 0.832 correctly specified, versus 0.766, 0.808, and 0.826 at assumed values of 0.25, 0.5, and 2.0 weeks. The worst case, a fourfold under-assumption, costs 6.6 points and still leaves it above both alternatives (0.746, 0.742). At a true half-life of 0.5 weeks it attains 0.934 correctly specified, against 0.964, 0.914, and 0.896 at the same four assumed values. Because Unadjusted and Lag-adjusted cannot depend on $\hat{t}_{1/2}$, their variation along that axis is pure Monte Carlo error, ranging 0.910 to 0.960 at a true half-life of 0.5 weeks. This roughly four- to five-point noise floor explains the apparent anomaly that assuming 0.25 weeks (0.964) beats correct specification (0.934), a difference well inside it. It also qualifies the one case where the direction is worth noting. At a true half-life of 0.5 weeks with an assumed value of 2.0 weeks, Exposure-weighted falls to 0.896, below both alternatives (0.926, 0.930), so a fourfold over-assumption erodes its advantage to a tie.

The practical reading is that an approximate half-life is adequate, with an asymmetry. Under-assuming is safer than over-assuming. This follows from the limiting-case relationship of Section 2.4, since shrinking $\hat{t}_{1/2}$ drives the predictor toward the binary indicator, which remains serviceable, whereas inflating it drives the predictor toward a constant, at which the on-off contrast and the estimand itself degenerate.

3.7.3 S3: dropout

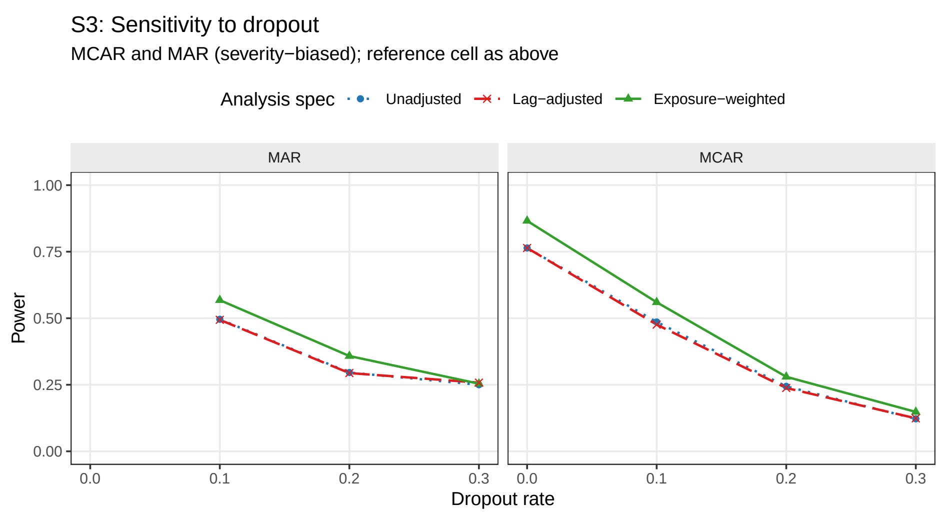

Figure 9: Power at dropout rates $\in \{0, 0.1, 0.2, 0.3\}$ under MCAR and MAR mechanisms.

All three specifications lose power monotonically as dropout increases. At the MCAR baseline ($N = 70$, no dropout), Exposure-weighted gives 0.866 ± 0.015 against 0.764 ± 0.019 for the other two. At 30% MCAR dropout, all three collapse to 0.122 to 0.148. The Exposure-weighted advantage narrows steadily as dropout worsens, from about 0.10 absolute at zero dropout to 0.07 at 10%, 0.04 at 20%, and 0.03 at 30% under MCAR, narrowing further under MAR at the highest rate. This is consistent with the advantage originating in the exposure-weighted contrast from off-drug observations, which contracts as those observations are preferentially lost. The advantage is preserved at dropout of 20% or below. Lag-adjusted again tracks Unadjusted throughout, the largest discrepancy being 0.010.

3.7.4 S4: biomarker-moderation effect-size curve

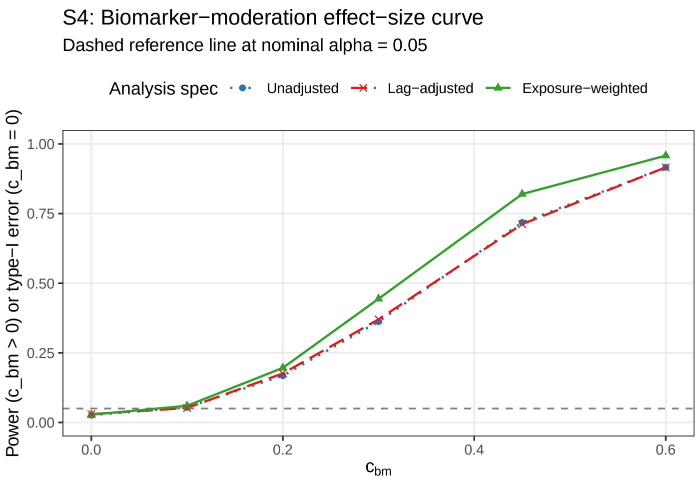

Figure 10: Power across the biomarker-moderation effect-size range $c_{bm} \in \{0, 0.10, 0.20, 0.30, 0.45, 0.60\}$. The $c_{bm} = 0$ point is the null for Type I verification.

The power curve is sigmoidal for all three specifications. At $c_{bm} = 0$, empirical Type I error sits below nominal for each (0.026 to 0.030). Power then rises through the mid-range and saturates once $c_{bm} \geq 0.45$, reaching 0.916 to 0.958 at $c_{bm} = 0.6$. The Exposure-weighted advantage over the unadjusted benchmark is largest at intermediate effect sizes, +0.082 absolute at $c_{bm} = 0.30$ and +0.102 at $c_{bm} = 0.45$, shrinking to +0.042 at $c_{bm} = 0.60$ as all three approach the ceiling, so the advantage matters most for trials targeting a moderate biomarker-treatment interaction. Lag-adjusted and Unadjusted coincide across the entire range, differing by at most 0.008.

3.7.5 S7: biomarker-interacted carryover terms

Every carryover term evaluated so far, Lag-adjusted (Section 2.4) and the exploratory washout-adjusted check of Section 4.1, enters the model as a nuisance main effect only, so by construction neither can repair the attenuated bm:D_{it} coefficient (Section 1). Here we test the two elaborations Section 4.5 identifies as the natural next step: crossing the same lagged and washout carryover terms with bm, so the biomarker effect itself is allowed to decline across the washout rather than committing to a rate, `E5` (Sx $\sim bm + t + D_{it} + \text{bm:}D_{it} + L_{it} + \text{bm:}L_{it}$) and `E6` (Sx $\sim bm + t + D_{it} + \text{bm:}D_{it} + t_{sd} + \text{bm:}t_{sd}$). Because these add a second interaction coefficient, the interaction test is not the same single-coefficient Wald statistic used elsewhere. We report the joint 2-df Wald test of bm:D_{it} and the second coefficient together (`nlme::anova.lme(Terms = ...)`), the only test that reflects whether coupling the biomarker to the carryover term recovers power, against Unadjusted and the washout main-effect check (`E4`) as single-coefficient benchmarks.

Table 4: Block S7: biomarker-interacted carryover terms at the reference cell (Hybrid, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$, exponential DGP, $n_{sim} = 500$, all replicates converged). Power for E5/E6 is the joint 2-df Wald test of bm:Db and the second interaction coefficient together; power for E1/E4 is the usual single-coefficient test. Mean bm:Db is the average point estimate of the bm:Db coefficient alone in each fitted model (comparable in form, not necessarily in calibration, across rows). The last column is the rejection rate of the bm:Db coefficient tested by itself, ignoring the second term, shown as a diagnostic only.

Specification	Test	Power (MCSE)	Mean bm:Db	bm:Db alone, reject. rate
Unadjusted	Single-coef.	0.748 (0.019)	-2.392	0.748
Washout main effect	Single-coef.	0.750 (0.019)	-2.394	0.750
Lag × bm	Joint (2 df)	0.690 (0.021)	-2.990	0.720
Washout × bm	Joint (2 df)	0.704 (0.020)	-1.561	0.198

Table 4 gives a clean negative result. Both interacted specifications have *lower* joint-test power than the unadjusted benchmark, `E5` by 5.8 points and `E6` by 4.4 points, despite each visibly moving the bm:D_{it} point estimate relative to `E1`. `E5`'s estimate is less attenuated on average (-2.990 against -2.392 for `E1`), consistent with the lagged interaction term absorbing some of the attenuation as intended, but the extra degree of freedom the joint test spends is not informative enough to turn that into a net power gain. `E6` is worse: its bm:D_{it} estimate is pulled toward zero (-1.561) and its single-coefficient rejection rate collapses to 0.198 from `E1`'s 0.748, evidence of severe coefficient instability from collinearity between bm:D_{it} and bm:t_{sd}, since t_{sd} is a near-deterministic function of D_{it} and elapsed time within this trial design. Interacting a carryover nuisance term with the biomarker is a theoretically motivated fix for the attenuation mechanism identified in Section 1, but at this reference cell it does not recover power over the unadjusted benchmark under either functional form, and the washout form additionally destabilizes the very coefficient it was meant to unbias. This closes the "adjacent specification… not evaluated" gap raised in Section 4.5, with a negative answer; it was checked at a single reference cell only and should not be read as ruling out every biomarker-interacted carryover term.

3.7.6 S8: architecture sensitivity of the specification ranking

Every result reported to this point, Tier 1 and blocks S1-S7 alike, is restricted to Architecture B (MVN differential correlation), the manuscript's stated scope (Section 2.3). Block S8 asks a narrower question than a full architecture comparison, which is the purpose of the companion manuscript on DGP architectures[3]: does the ranking among analysis specifications established above under Architecture B hold under Architecture A (mean moderation) as well, or is the choice of carryover representation itself architecture-dependent? We refit the five specifications of block S7's table (`E1`, Lag-adjusted `E3`, Exposure-weighted `E2`, `E5`, `E6`) at the same reference cell under both architectures. Because c_{bm} is not calibrated to equal power across architectures, the two produce materially different absolute power at the same nominal value (Section 1;[3] shows the mean-moderation channel requires several times the nominal effect size for a comparable power loss under carryover); only the within-architecture ranking is the object of interest here.

Table 5: Block S8: analysis-specification ranking by DGP architecture, at the reference cell (Hybrid, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$, exponential DGP, $n_{sim} = 500$). E5/E6 power is the joint 2-df Wald test as in Table
reftab:tab-s7. Absolute power is not comparable between architectures at the same nominal c_{bm}; only the within-architecture ranking is of interest.

Architecture	Specification	Power (MCSE)	Non-conv
B (MVN)	Unadjusted	0.748 (0.019)	0.000
B (MVN)	Lag-adjusted	0.748 (0.019)	0.000
B (MVN)	Exposure-weighted	0.826 (0.017)	0.000
B (MVN)	Lag $\times$ bm	0.690 (0.021)	0.000
B (MVN)	Washout $\times$ bm	0.704 (0.020)	0.000
A (mean mod.)	Unadjusted	0.958 (0.009)	0.000
A (mean mod.)	Lag-adjusted	0.958 (0.009)	0.000
A (mean mod.)	Exposure-weighted	0.946 (0.010)	0.000
A (mean mod.)	Lag $\times$ bm	0.918 (0.012)	0.000
A (mean mod.)	Washout $\times$ bm	0.924 (0.012)	0.000

Table 5 shows the ranking is architecture-dependent for exactly one specification. Under Architecture B, Exposure-weighted leads by 7.8 points (0.826 versus 0.748 for Unadjusted/Lag-adjusted); under Architecture A the advantage does not merely shrink but reverses in direction, Exposure-weighted trailing marginally (0.946 versus 0.958, a 1.2-point gap within one MCSE, so not distinguishable from chance on this evidence alone). This matches the pattern already reported from the full-scope grid in the archived drafts, that Exposure-weighted is marginally dominated in every design examined under mean moderation (Section 4.6), now confirmed directly within this manuscript at its own reference cell rather than resting on the archived material alone. Lag-adjusted (`E3`) remains statistically indistinguishable from Unadjusted (`E1`) under both architectures (0.748 versus 0.748 under B, 0.958 versus 0.958 under A), extending Section 3.3's clean null to Architecture A as well. The S7 negative result also replicates here: `E5` and `E6` sit below the `E1`/`E3` benchmark under both architectures (0.918 and 0.924 against 0.958 under A; 0.690 and 0.704 against 0.748 under B), so biomarker-interacted carryover terms recover power in neither architecture examined. The one specification recommendation this manuscript makes, exposure weighting in the Hybrid and OL+BDC designs (Conclusions), is therefore itself conditional on Architecture B; this check confirms that restriction rather than merely asserting it, and the recommendation should not be extrapolated to Architecture A, where the archived full-scope drafts remain the authority.

3.8 A higher-precision check

Because a number of the Tier 1 gaps are small, we reran a focused 24-cell grid (OL+BDC design, $N \in \{35, 70\}$, true half-life $\in \{0.5, 1.0\}$) at 600 trials per cell, all converging, and compared specifications with a paired McNemar test, considerably more sensitive than an independent-groups comparison since all three are fit to the same trials. Type I error across the twelve null cells fell in $[0.003, 0.053]$, so nominal alpha holds. Paired McNemar tests place

Exposure-weighted ahead of the unadjusted benchmark at all four alternative cells, by +2.3 to +13.7 percentage points, with the largest gap at the highest-leverage cell ($N = 70$, true $t_{1/2} = 1.0$ weeks, $p = 6.6\times10^{-17}$) and $p \leq 0.015$ at the remaining three, a sharper confirmation than the Tier 1 grid alone provides.

One scope caveat governs the rerun's third arm. It was produced by the package's `lme_analysis()` routine, whose carryover option appends time since discontinuation as a linear main effect rather than the lagged indicator L_{it} used in the Tier 1 grid. It is therefore not the Lag-adjusted specification of Section 2.4 and is not reported as such here. That alternative remains a reasonable specification we have not evaluated (Section 4.5).

4 Discussion

4.1 The lagged-treatment term does no work

The clearest result of this study is a negative one. Adding the classical lagged-treatment indicator to the analysis model changed nothing measurable, anywhere in the grid. Across all 216 cells the mean power difference against the unadjusted benchmark was 0.003 against an MCSE of 0.018, bias, MSE, and coverage agreed to three decimal places, and the equality held in every sensitivity block and all five cluster-robust stress cells under both inference procedures. The strength of this conclusion rests on the design rather than the grid size. The two specifications estimate the same coefficient, were fitted to the same simulated datasets under common random numbers, and differ by exactly one term, so there is no confounding from a change of estimand, Monte Carlo variation between arms, or differential test size (both are conservative to the same degree, Section 3.5). What is left is the term itself, and it contributed nothing.

The mechanism follows from where the term sits. Carryover damages the interaction test on two fronts (Section 1). It attenuates the coefficient, because contaminated occasions are coded as drug-free, and it inflates the residual variance, because those occasions are heterogeneous in a way the model does not represent. The lagged indicator enters only as a nuisance main effect, with no bm:L_{it} product term, so the tested interaction remains bm:D_{it} and the attenuation is untouched. It can in principle absorb some of the second effect, but evidently absorbs too little to matter, since it marks a single occasion per randomization path and is constant thereafter, unable to describe a graded decline across a two- to four-occasion washout.

A more capable, elapsed-time-indexed nuisance term might succeed where the crude lagged indicator fails, so we ran a targeted check on the alternative the reference implementation provides, `Sx ~ bm + t + Db + bm:Db + tsd`, called the washout-adjusted specification (`E4`), across the three designs at true half-lives of 0.5 and 1.0 weeks ($N = 70$, $c_{bm} = 0.45$, exponential DGP, 500 replicates). It does slightly better, and the improvement is instructive rather than material. Type I error is at nominal for both (mean 0.050 versus 0.045), and power exceeds the benchmark in all six cells by a mean of 1.3 points (sign test $p = 0.031$), reaching significance under crossover at both half-lives (+1.8, $p = 0.008$; +2.2, $p = 0.003$) and OL+BDC at the longer one (+2.2, $p = 0.015$), but not Hybrid (+1.0, $p = 0.27$). The mechanism is exactly the one Section 2.4 predicts. Point estimates are unchanged, so nothing about the attenuation is repaired, and the entire effect runs through the standard error, which only a long enough washout gives it something to describe. The crossover schedule places off-drug occasions 2.5 to 10 weeks after discontinuation, against one to two weeks for Hybrid, so the practical ceiling on what any carryover nuisance term can deliver is about two percentage points, and only in long-washout designs, against the nine to eleven points exposure weighting delivers where it works. This check covers only the exponential DGP at a single N and effect size, so it is a probe rather than a fourth arm, and shares every specification's structural limitation of entering carryover as a main effect only (Section 4.5). For practitioners the implication is direct. A carryover nuisance term of either form should not be expected to recover meaningful power lost to carryover, and reporting one does not constitute an effective adjustment for it.

4.2 The exposure-weighted advantage, and where it fails

The exposure-weighted predictor does help, but conditionally. At its best cell (Hybrid, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$) it exceeds the unadjusted benchmark by about 10 points of power (0.860 versus 0.766, roughly five MCSEs and so not attributable to chance), confirmed at all four cells of the Section 3.7 high-precision rerun. Under the crossover design it performs considerably worse (0.488 versus 0.830), apparently because the within-subject correlation there already carries most of the signal the decaying predictor was designed to recover, so it gains little while its down-weighting of off-drug observations works against it. The recommendation therefore holds only in the two non-crossover designs, and should not be extended to a crossover trial, where the unadjusted binary indicator is materially better. That the one specification addressing attenuation is also the only one to gain power, while the one addressing only residual variance gains nothing, suggests attenuation is the mechanism that matters here, an interpretation consistent with the results rather than a finding the design isolates directly.

4.3 What the robustness checks add

The Section 3.6 sensitivity blocks show the exposure-weighted advantage, where it exists, is not fragile. It survives incorrect decay-shape assumptions (Figure 4), an incorrectly assumed half-life short of a large over-assumption at a short true half-life (Figure 8), and holds its five- to eight-point lead across the autocorrelation range examined (Figure 7). One caveat qualifies the Tier 1 figures throughout. Outside block S2 the analysis-side half-life is set equal to the data-generating one, granting Exposure-weighted an oracle no real analyst possesses, so its Tier 1 margins are upper bounds and block S2 gives the more realistic figure.

4.4 Dropout matters more than the method

The finding with the largest practical consequence concerns missing data, not carryover. When patients drop out (Figure 9), power collapses for all three specifications together, to between 0.12 and 0.25 at 30% dropout, essentially indistinguishable. No specification offers protection, since each can only organize whatever data remain. MAR dropout is somewhat worse than MCAR, because the sickest patients, who carry the most signal, tend to leave first. Retaining patients in the trial yields considerably more power than any choice among these three specifications ever could.

4.5 Scope, related work, and limits

This manuscript deliberately narrows the full-scope grid to the covariance-moderation architecture and to the non-linear decay forms, in the interest of a self-contained evaluation of manageable length. The archived full-scope drafts (`archive/report.Rmd` and `archive/report-slim.Rmd`) show that the exposure-weighted advantage documented here does not carry over to mean moderation, where the biomarker moderates the mean response directly and Exposure-weighted is marginally dominated in every design examined. Readers who need that cross-architecture comparison should consult those drafts rather than extrapolate from this one. Establishing what the originally published power values (Section 3.2) were computed under remains outstanding work.

The crossover tradition surveyed by Jones and Kenward[4] would lead one to expect a shape-free lagged term to outperform a parametric predictor once the decay shape is mis-specified. Our results do not support that expectation over the range examined, for two reasons. The exposure-weighted penalty for an incorrect half-life is too small to be overtaken (Section 3.6), and more fundamentally the lagged term does not improve on ignoring carryover at all (Section 4.1), leaving no advantage for decay-shape mis-specification to erode.

One adjacent specification bounds the negative result of Section 4.1. Every carryover term considered there, lagged and washout-adjusted alike, enters as a nuisance main effect with no product against the biomarker, so that finding is that a carryover *main effect* does not recover the attenuated signal, not that no shape-free specification can. The natural candidate is a carryover term interacted with the biomarker, either `Sx ~ bm + t + Db + bm:Db + L + bm:L` or its washout-adjusted counterpart with `bm:tsd`, letting the biomarker effect decline across the washout without committing to a rate. Block S7 (Section 3.6) tests exactly these two specifications and finds the same negative result extends to them: at the reference cell, both give lower power than the unadjusted benchmark once evaluated by the joint test their extra coefficient requires, and the washout form additionally destabilizes the bm:D_{it} estimate through collinearity with bm:t_{sd}. Block S8 further shows this negative result, and the Lag-adjusted null of Section 3.3, both replicate under Architecture A, so they are not an artifact specific to covariance moderation. Only the Exposure-weighted advantage is architecture-sensitive, reversing in direction, though not significantly, under Architecture A (Section 3.6, Block S8).

Two further limitations bound what we can conclude. The numbers presented here come from a single parameter set calibrated to a prazosin and PTSD trajectory, so absolute power values should be read as illustrative and the ranking, itself conditional on trial design, as the more portable result. And the sensitivity checks vary one factor at a time and can therefore miss genuine interactions between factors, for instance between high autocorrelation and heavy dropout. The one joint check available to us, block S5 in the full-scope manuscript, revealed no such interaction.

5 Conclusions

1. **The lagged-treatment term does nothing.** It never improved on ignoring carryover altogether, on any measure, in any design, at any half-life, and in every sensitivity block, a clean null given the two specifications share an estimand and differ by a single term. The term enters as a nuisance main effect only, leaving the attenuation of the interaction untouched, and should not be reported as an adjustment for carryover.

2. **Exposure weighting helps, but conditionally.** It achieves the highest power, about 10 points over the unadjusted benchmark, only in the Hybrid and OL+BDC designs, and its lead survives mis-specified decay shape and half-life there. Under crossover it is markedly worse than the binary indicator (0.488 against 0.830) and should not be used. Where used, we recommend a CR2 cluster-robust standard error (Block S6).

3. **An approximate half-life is adequate, but err short.** A fourfold error at a true half-life of one week costs 6.6 points of power, so a rough pharmacokinetic estimate suffices in practice. The error is asymmetric. Under-assuming drives the predictor toward the still-serviceable binary indicator, while over-assuming erodes the advantage to a tie. Outside block S2 the analyst is granted the true half-life, so the Tier 1 margins are upper bounds.

4. **Dropout matters more than the choice of method.** Power falls sharply as dropout increases, roughly halving at 20%, equally across all three specifications. Preventing dropout is a more effective use of design effort than any choice among them.

5. **The interaction test is somewhat conservative, and correctably so.** All three specifications reject slightly less often than nominal under the null. A CR2 cluster-robust standard error restores the correct rate while recovering modest power, and the exposure-weighted advantage is held or widened under CR2 at four of five S6 stress cells (Block S6).

6. **This is a narrowed slice of a larger study.** Mean moderation and linear decay fall outside this manuscript's scope. The archived drafts `archive/report.Rmd` and `archive/report-slim.Rmd` are the authority on those comparisons and should be consulted before generalizing beyond covariance moderation. We also make no claim of numerical agreement with the originally published power values (Section 3.2).

7. **Biomarker-interacted carryover terms do not help either, and the finding is not architecture-specific.** Crossing a lagged or washout carryover term with the biomarker, the natural elaboration of the negative result in item 1, gives lower power than the unadjusted benchmark once evaluated by the joint test its extra coefficient requires (Block S7), and the washout form additionally destabilizes the interaction coefficient it was meant to unbias. Block S8 confirms this negative result, and the Lag-adjusted null of item 1, both hold under Architecture A as well as B. Only the Exposure-weighted advantage is architecture-sensitive: it reverses in direction, though not distinguishably from chance at this cell, under Architecture A, confirming that item 2's recommendation is conditional on Architecture B and should not be extrapolated to

Architecture A.

6 Conflict of interest

The authors declare no conflicts of interest.

7 Funding

This work received no specific funding.

8 Data availability statement

The data and code supporting the findings of this study are available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`. Simulation drivers are at `analysis/scripts/carryover-sensitivity/`. Exact paths for this manuscript are listed in the Reproducibility section below.

9 Reproducibility

This manuscript was rendered with R 4.6.0 inside the project's zzcollab Docker container. The production simulation ran under R 4.5.3 (§3.6), with the image hash in `Dockerfile` and the package set in `renv.lock`. The simulation study uses `nlme` for the continuous-time linear-mixed analysis, `parallel` for the 8-core parameter-grid sweep, and `data.table` or `furrr`/`purrr` in the two simulation collections. No simulation code runs during knitting, which loads pre-computed checkpoints only. Each driver sets a master seed (20260507) and derives per-replicate seeds from it via `parallel::mc.set.seed`.

Data and code availability. All simulation outputs, cell-level summaries, and driver scripts are versioned under `analysis/scripts/carryover-sensitivity/` and `analysis/data/` in the pmsimstats-ng repository, including the principal factorial, the S1 to S5 sensitivity blocks, the S6 cluster-robust recalibration, the washout-adjusted (E4) check of Section 4.1, the high-precision prototype rerun of Section 3.7, the S7 biomarker-interacted carryover terms block (`09-run-sensitivity-s7.R`, `output/09-sensitivity-s7.rds`), and the S8 architecture-sensitivity block (`10-run-sensitivity-s8.R`, `output/10-sensitivity-s8.rds`), both added 2026-08-18 (master seed 20260415, 500 replicates each, reference cell only). The five- and six-specification fitters for S7/S8 (`fit_spec_s7`, `simulate_cell_s7`) are appended to `simulation-core.R` alongside the Tier 1 `fit_spec`/`fit_all_specs` pair, which they do not modify. Figures 1/2 are produced from the same `02-grid-summary.rds` Tier 1 summary as the headline table, by `12-hendrickson-heatmap.R`. The grid design and Morris ADEMP pre-registration are in `simulation-grid-plan.md` in the same directory. Superseded wider-scope drafts are retained under `archive/`, documented in its `README.md`.

Specification labels. The three specifications are stored as `E1`, `E2`, `E3` and displayed here as Unadjusted, Lag-adjusted, and Exposure-weighted. The mapping is applied only at display time, by `spec-labels.R`, so the stored values and archived outputs are unaffected. Output archived before 2026-07-31 stores the specifications as `A1`, `A2`, `A3`. The migration is documented in `analysis/report/NOTATION.md`.

10 References

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